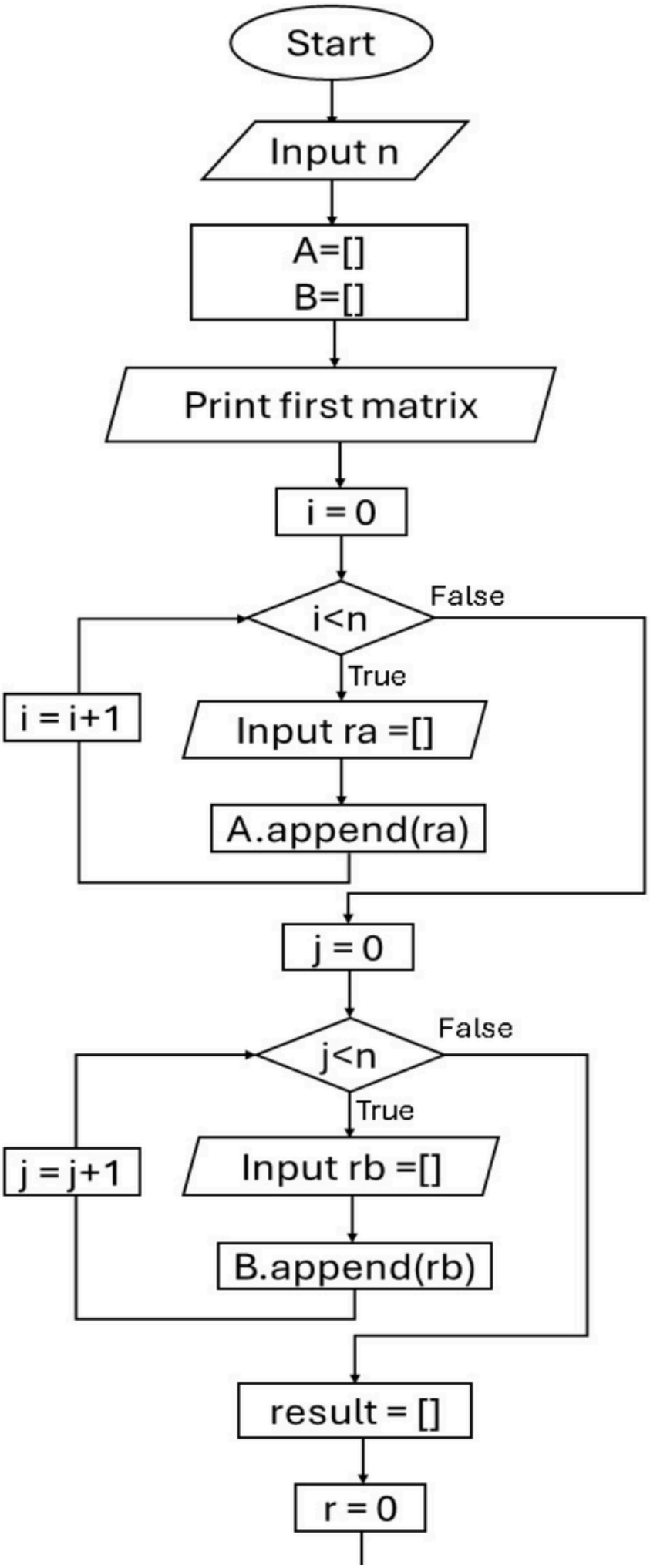
Experiment 8

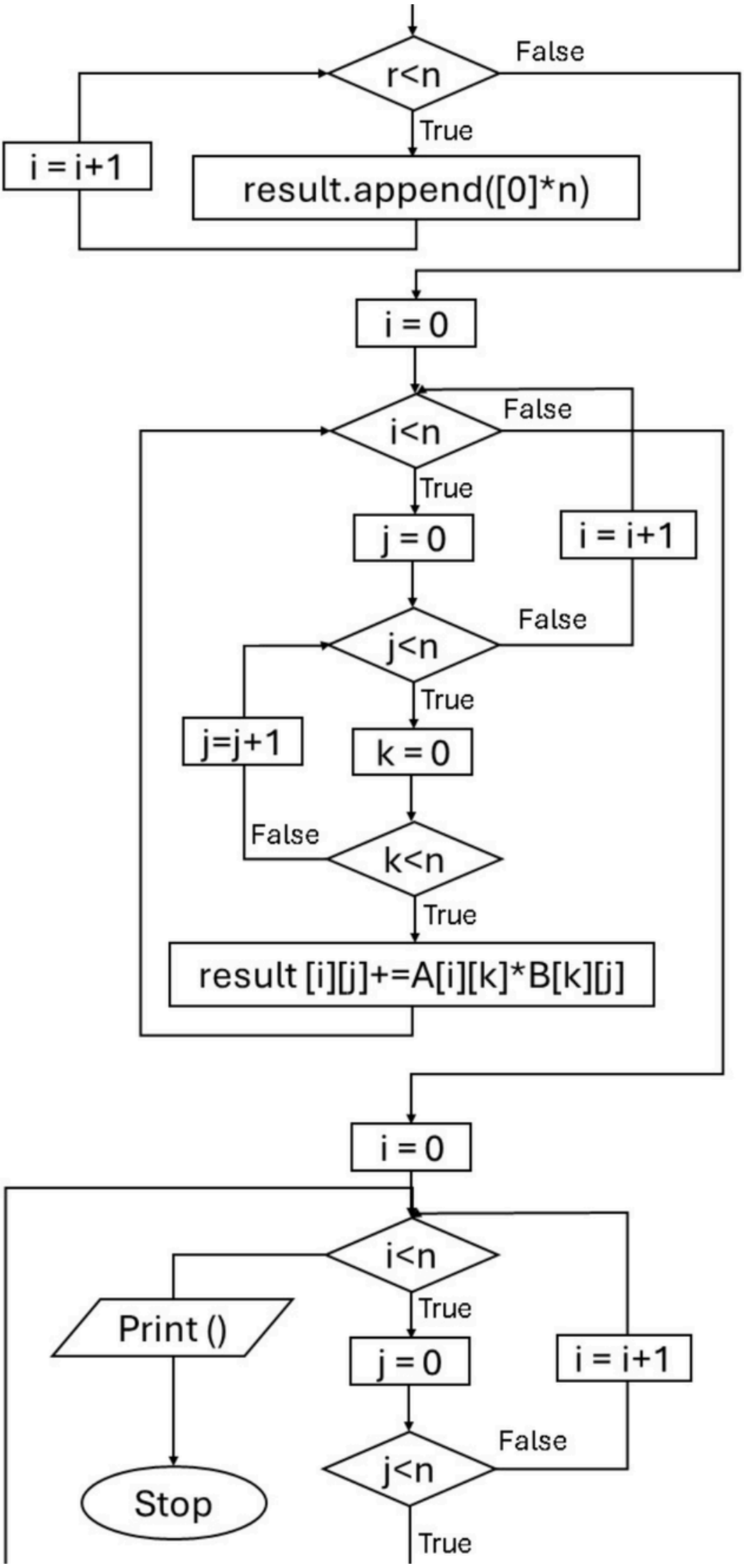
8.1.1 Multiplication of Two Square Matrix

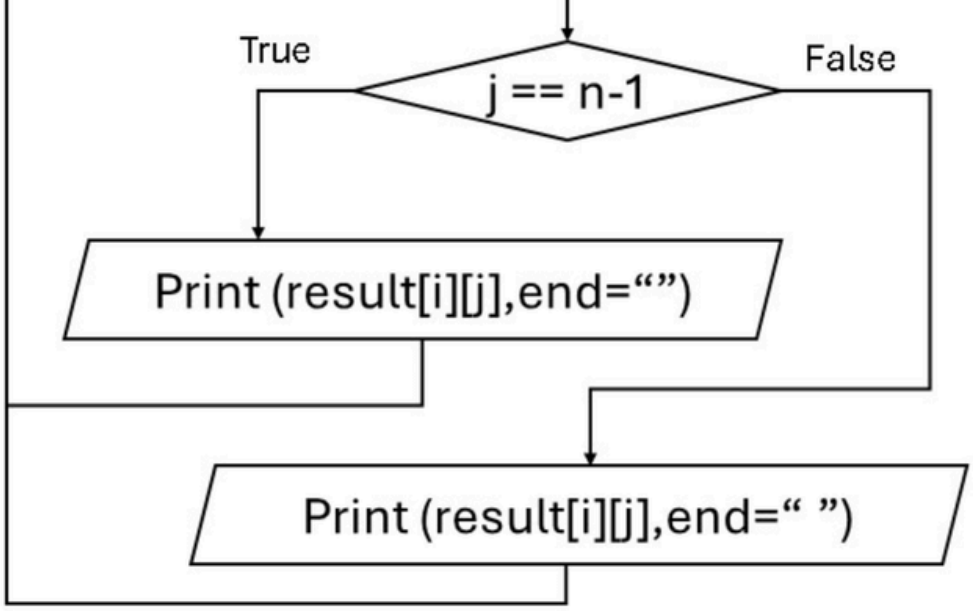
Algorithm :

```
Step 1: Start
Step 2: Input Dimension (n)
Step 3: A=[]
 Create an empty list A
 Set loop a in range 0 to n:
        Input a row of integers and append to A
Step 4: B=[]
 Create an empty list B
 Set loop b in range 0 to n:
        Input a row of integers and append to B
Step 5: Result =[]
 Create an empty list result
 Set loop r in range 0 to n:
        Append a list of n zeros to result
Step 6: Multiplication of Matrix
 Set loop i in range 0 to n:
        Set loop j in range 0 to n :
              Set loop k in range 0 to n :
                    result[i][j] += (A[i][k] * B[k][j])
Step 7: Print Resultant Matrix
 Set loop i in range $0$ to $n$:
        Set loop j in range $0$ to $n$:
              If j is the last index :
                    Print result[i][j] (end="")
              Else:
                    Print result[i][j](end=" ")
       Print a new line.
Step 8: Stop
```

Flowchart :


Start
Input n
A=[]
B=[]
Print first matrix
i = 0
i<n
False
True
i = i+1
Input ra =[]
A.append(ra)
j = 0
j<n
False
True
j = j+1
Input rb =[]
B.append(rb)
result = []
r = 0


r<n
False
True
i = i+1
result.append([0]*n)
i = 0
i<n
False
True
j = 0
i = i+1
j<n
False
True
j=j+1
k = 0
False
k<n
True
result [i][j]+=A[i][k]*B[k][j]
i = 0
i<n
True
Print ()
j = 0
i = i+1
j<n
False
Stop
True



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8.1.1. Multiplication of Two Square Matri... 04:46

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices $(n \times n)$.
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases

Explorer

Multiplica... Submit

```
n = int(input("dimension: "))
print("first matrix:")
matrix1 = []
for i in range(n):
    row = list(map(int, input().split()))
```

Average time **0.059 s** 59.00 ms

Maximum time **0.065 s** 65.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 48 ms Debug

Expected output	Actual output
dimension: 2	dimension: 2
first matrix:	first matrix:
1 2	1 2

Debugger

Prev Reset Submit Next

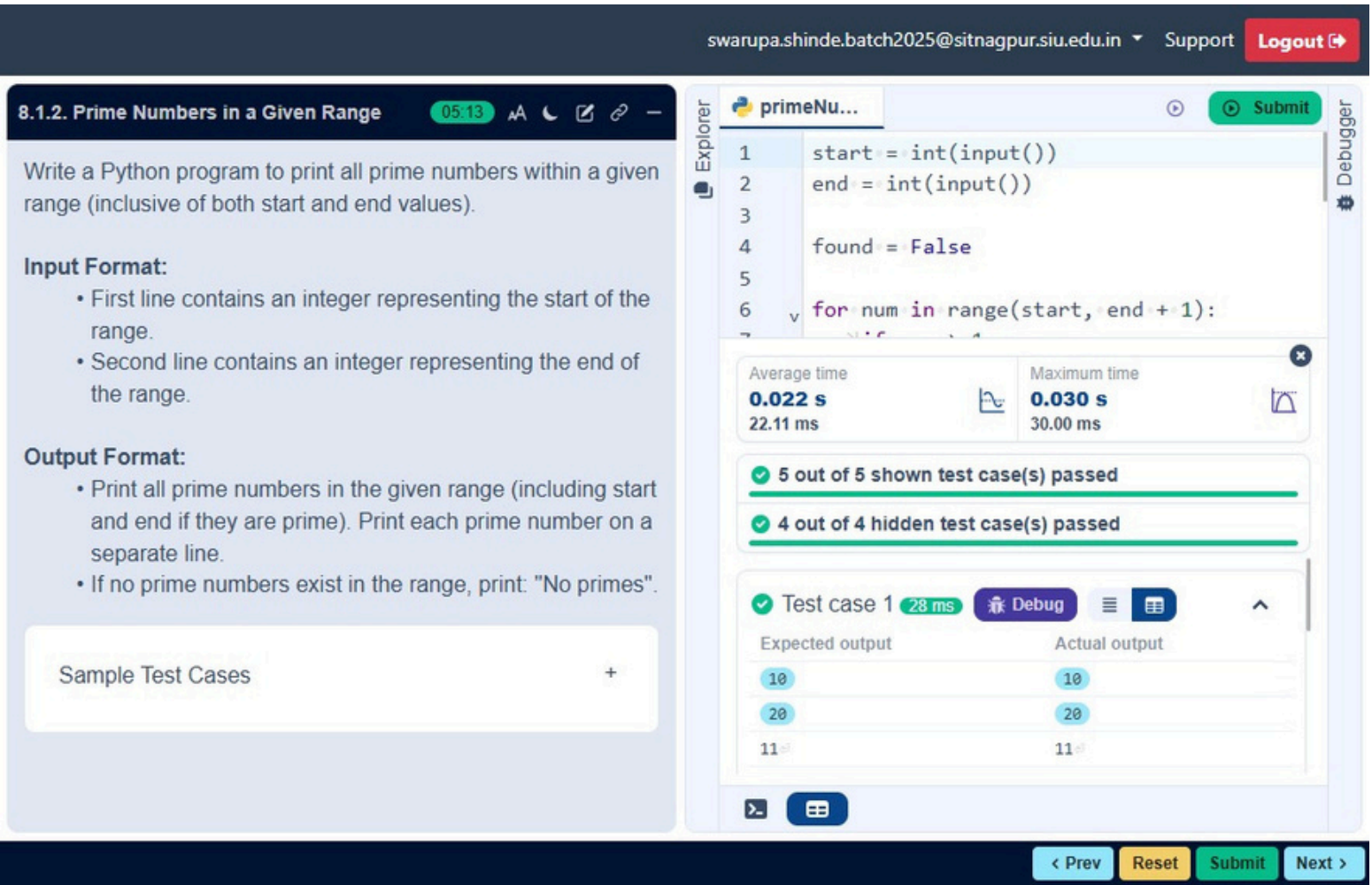
Experiment 8

8.1.2 Prime Numbers in a Given Range

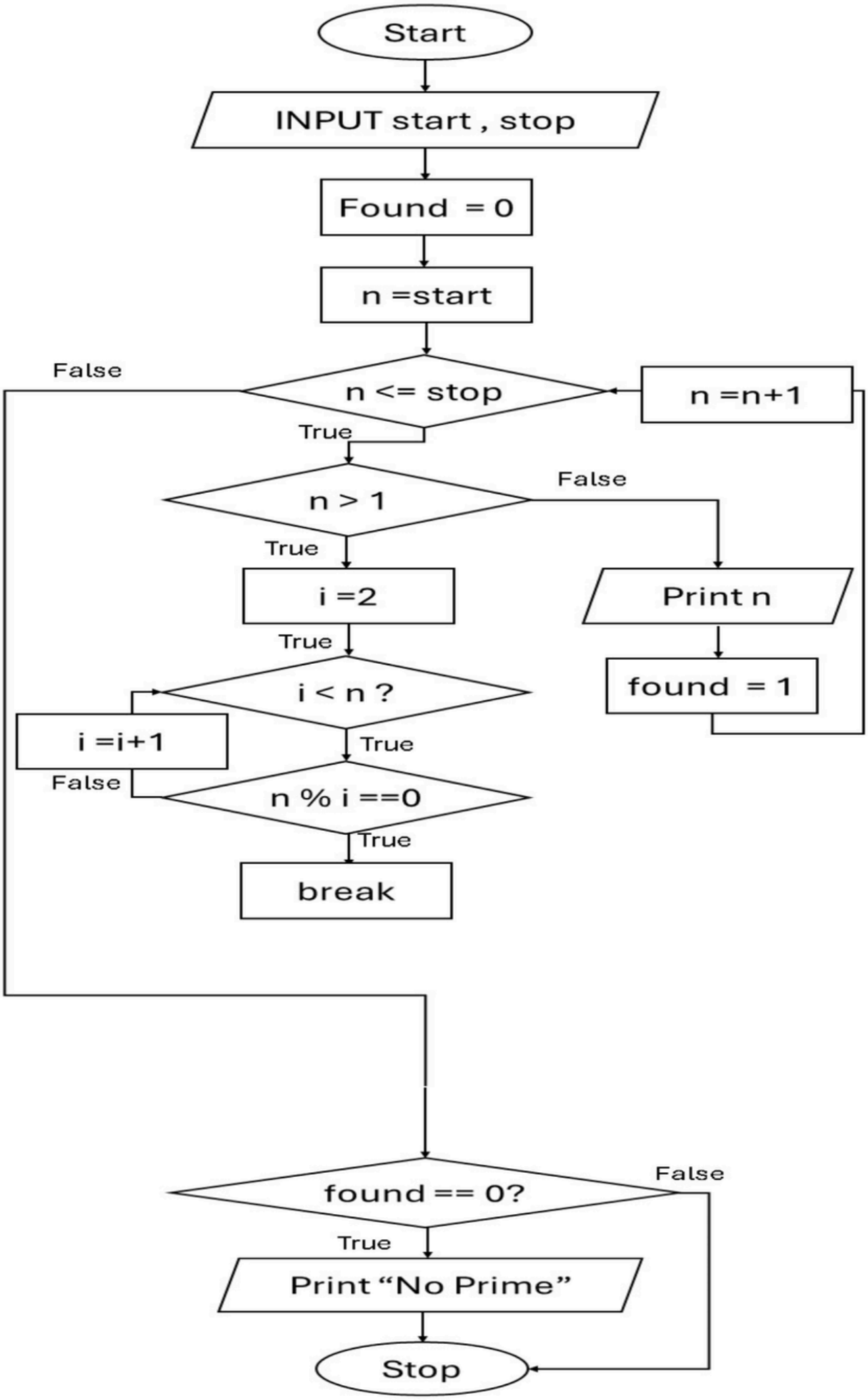
Algorithm :

Step 1: Start
Step 2: Input Start , Stop
Step 3: found = 0
Step 4: Set for loop n in range (start , stop+1)
 if n > 1:
  Set for loop I in range (2, n):
   if n % i==0:
    then n is not prime
   break the loop
  else :
   n is prime
   found =1
   print n

Step 5 : If found == 0 :
  i.e n is not prime :
  print not prime
Step 6: Stop



Flowchart :


Start
INPUT start , stop
Found = 0
n =start
False
n <= stop
n =n+1
True
n > 1
False
True
i =2
Print n
True
i < n ?
found = 1
i =i+1
True
False
n % i ==0
True
break
found == 0?
False
True
Print "No Prime"
Stop